Solution to Exercise 5A: Another Perspective on Regularisation

(Based on a solution written up by Hans Martin Sandager)

a

Substituting we get

$$\begin{aligned}\arg\max_\theta \log f(\theta|\mathbf{x}) &= \arg\max_\theta \log \frac{f(\mathbf{x}|\theta)f(\theta)}{f(\mathbf{x})} \\ &= \arg\max_\theta \log f(\mathbf{x}|\theta) + \log f(\theta) - \log f(\mathbf{x})\end{aligned}$$

$f(\mathbf{x})$ is a constant, and does not affect the maximisation. Using that f is a Gaussian pdf with $\mu = 0$ we get

$$\log f(\theta) = -\frac{\log 2\pi\sigma_\theta^2}{2} - \frac{\theta^2}{2\sigma_\theta^2}$$

The first term doesn't affect the maximisation since it is constant, and we are left with only $\log f(\mathbf{x}|\theta)$ which is a log likelihood function and $-\frac{\theta^2}{2\sigma_\theta^2}$

$$\arg\max_\theta \log f(\theta|\mathbf{x}) = \arg\max_\theta \log f(\mathbf{x}|\theta) - \frac{1}{2\sigma_\theta^2}\theta^2.$$

Normally the regularisation term is added to a loss function, but since the likelihood function $f(\mathbf{x}|\theta)$ is the opposite of a loss function, the regularisation term is subtracted instead. The regularisation hyper parameter is $\frac{1}{2\sigma_\theta^2}$

b

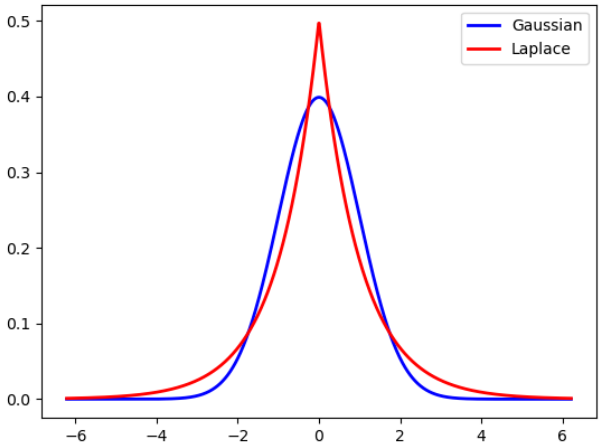
We use the same method as in the previous exercise but this time

$$\log f(\theta) = -\log 2b - \frac{|\theta|}{b}.$$

By removing $-\log 2b$, we see that this corresponds to ℓ_1 regularisation (because of the absolute value) and the hyper parameter is $\frac{1}{b}$

c

The regularisation will in both cases penalise the numerical size of θ, in the case of the Gaussian distribution model, the hyper parameter which penalises θ is inversely proportional to the variance, meaning that if the variance of the parameter distribution is low, high values of sigma are penalised harder than if σ_θ^2 was high. The same is the case when f is the Laplacian pdf, where b is a way of measuring the variance of the distribution is $2b^2$.



Compared to the Gaussian distribution of ℓ_2 regularisation, the ℓ_1 Laplace has a larger peak at 0, representing the higher probability of a weight being exactly 0. By contrast, its tails are heavier, so if a weight is non-zero, it also more likely to have a greater magnitude in the ℓ_1-regularised scenario. In other words, in ℓ_1 regularisation, few weights will be non-zero, but those that are can take on relatively large values, whereas ℓ_2 regularisation will tend to keep all weights small, but non-zero.